

Introduction to Newton's Laws

Newton's First Law.

*Slides adapted from Peggy Bertrand

Isaac Newton

- Arguably the greatest scientific genius ever.
- Came up with 3 Laws of Motion to explain the observations and analyses of Galileo and Johannes Kepler.
- Discovered that white light was composed of many colors all mixed together.
- Invented new mathematical techniques such as calculus and binomial expansion theorem in his study of physics.
- Published his Laws in 1687 in the book Mathematical Principles of Natural Philosophy.

Newton's First Law

- A body in motion stays in motion at constant velocity and a body at rest stays at rest unless acted upon by a net external force.
- This law is commonly referred to as the Law of Inertia.

The First Law is Counterintuitive

**Even today people
still believe that a
force is required to keep
an object moving.**

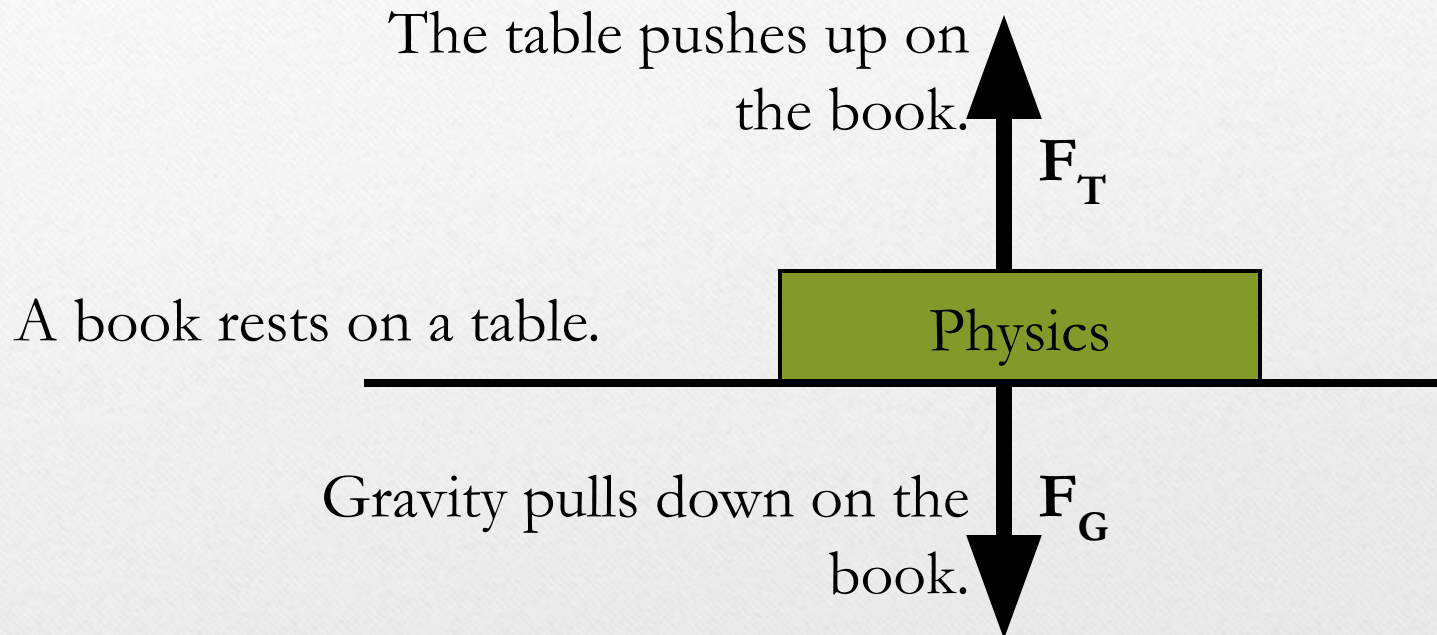


Aristotle firmly believed this.
But we know better!

Implications of Newton's 1st Law

- If there is **zero net force** on a body, it cannot accelerate, and therefore must move at constant velocity, which means
 - it cannot turn,
 - it cannot speed up,
 - it cannot slow down.

What is Zero Net Force?



Even though there are forces on the book, they are balanced.
Therefore, there is no **net force** on the book.

$$\Sigma F = 0$$

Diagrams

- Draw a **free body diagram** for a book sitting on a table.

Free Body Diagram



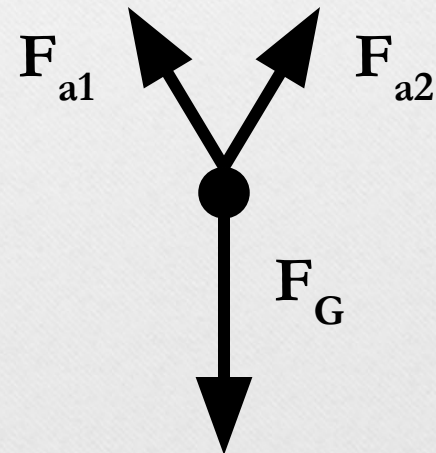
Sample Problem

- a) A monkey hangs by its tail from a tree branch. Draw a force diagram representing all forces on the monkey



Sample Problem

b) Now the monkey hangs by both hands from two vines. Each of the monkey's arms are at a 45° from the vertical. Draw a force diagram representing all forces on the monkey.

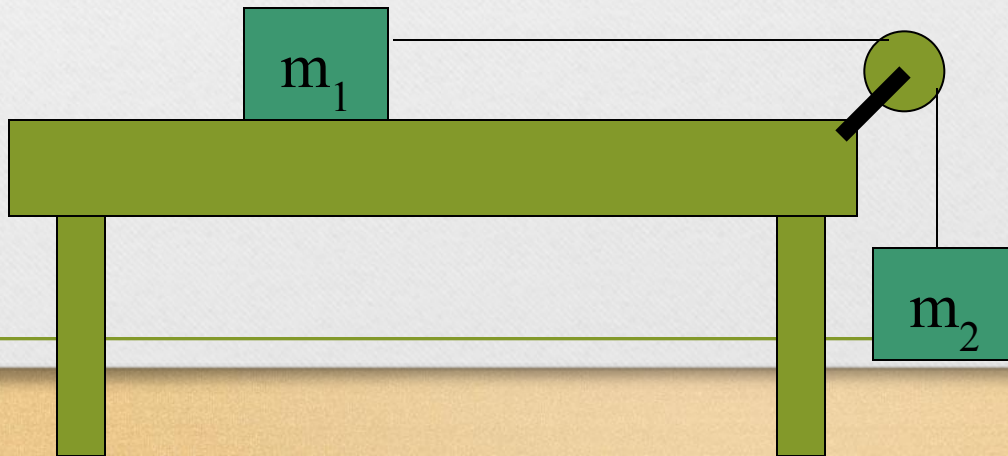


Sample problem

A 5.0-kg bag of potatoes sits on the bottom of a stationary shopping cart. Sketch a free-body diagram for the bag of potatoes. Now suppose the cart moves with a constant velocity. How does this affect the free-body diagram?

Sample problem

Mass 1 (10 kg) rests on a table connected by a string to Mass 2 (5 kg). Find the minimum coefficient of static friction for which the blocks remain stationary.



Sample problem - solution

$$\Sigma F = 0$$

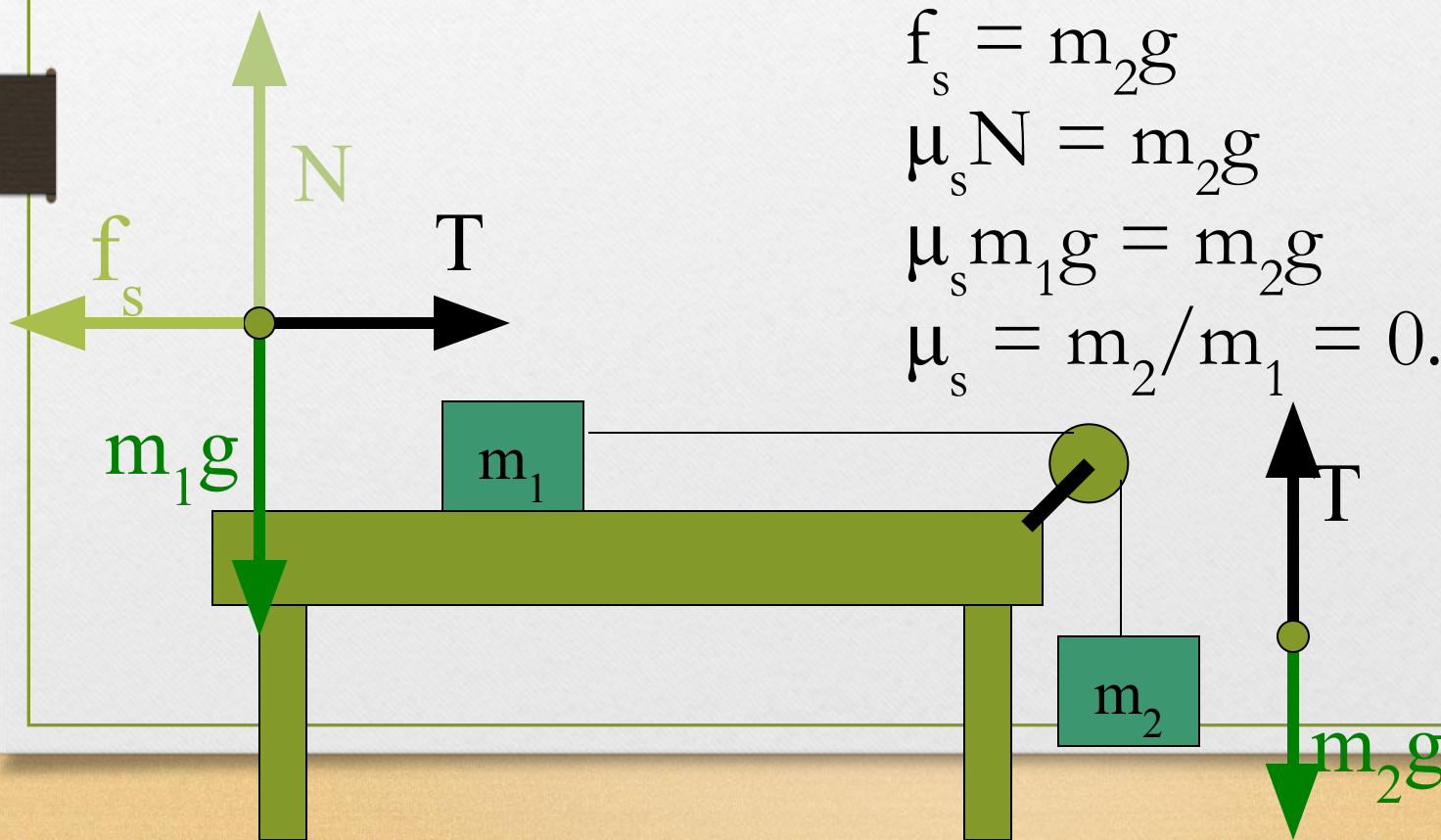
$$m_2 g - T + T - f_s = 0$$

$$f_s = m_2 g$$

$$\mu_s N = m_2 g$$

$$\mu_s m_1 g = m_2 g$$

$$\mu_s = m_2 / m_1 = 0.50$$



Working 1st Law Problems

1. Draw a force or free body diagram.
2. Set up 1st Law equations in each dimension.
$$\Sigma F_x = 0 \text{ and/or } \Sigma F_y = 0$$
3. Identify numerical data.
x-problem and/or y-problem
4. Solve the equations.
5. Substitute numbers into equations.
“plug-n-chug”

WARMUP

A 10-kg box is being pulled across the table to the right at a constant speed with a force of 50N.

- a) Calculate the Force of Friction
- b) Calculate the Force Normal
- c) Calculate the coefficient of kinetic friction

NFL cont'd

A person pushes on a 20 kg crate to the right with a 150 N force that is angled at 40° above the horizontal. The crate moves at a constant velocity of 5.2 m/s.

- a) Calculate the Force of Friction
- b) Calculate the Force Normal
- c) Calculate the coefficient of friction